



2IA Practice Problems III

Groups



Practice Problems Chapter 3

1. In each case determine whether the statement is true or false. **You will get zero marks if you do not provide an argument supporting your answer.**
 - (a) If H and K are two subgroups of a group G , then their union $H \cup K$ is also subgroup of G .
 - (b) The cyclic group $\mathbb{Z}_n$ has exactly n generators.
 - (c) Every abelian group is cyclic.
 - (d) The intersection of any two subgroups of a group G can never be the empty set.
 - (e) $\mathbb{Z}_2$ is a subgroup of $\mathbb{Z}_4$.
 - (f) Every two cyclic groups of the same order are isomorphic.
 - (g) There exists group G that is isomorphic to a proper subgroup of itself.

Solution:

(a) **False:** Take for example $G = \mathbb{Z}$ and $H = 2\mathbb{Z}$ and $K = 3\mathbb{Z}$. Then $H \cup K$ is not a subgroup of $\mathbb{Z}$, since $2, 3 \in H \cup K$ but $2 + 3 = 5 \notin H \cup K$, since it is not a multiple of 2 or 3.

(b) **False:** Consider, for example, $\mathbb{Z}_4 = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}\}$. Recall that $\bar{a}$ is a generator of $\mathbb{Z}_4$ if and only if $\gcd(a, 4) = 1$. From here, we obtain that $\mathbb{Z}_4$ has two generators, namely, $\bar{1}$ and $\bar{3}$.

(c) **False:** As seen in class the group $\mathbb{Z}_2 \times \mathbb{Z}_2$ is not cyclic (all its elements have order 2) and it is abelian.

(d) **True:** Every subgroup contains the identity element of G , so the intersection of any two subgroups contains at least the identity element.

(e) **False:** A subgroup (is by definition) a subset of a group such that it is a group itself with the same operation. In this case, $\mathbb{Z}_2$ is **NOT** a subset of $\mathbb{Z}_4$, and so it does not make sense to think about subgroups.

Be very careful here, we are using the same notation here to denote $\bar{0} \in \mathbb{Z}_2$ and $\bar{0} \in \mathbb{Z}_4$ but these are not the same! $\bar{0}$ viewed as an element of $\mathbb{Z}_2$ denotes the even numbers, while $\bar{0} \in \mathbb{Z}_4$ denotes all the integers that have a remainder of 0 when divided by 4; these two sets are not equal!

(f) **True:** This follows from the Isomorphism Theorem of Cyclic Groups: every cyclic group of order n is isomorphic to $\mathbb{Z}_n$, while every cyclic group of order infinite is isomorphic to $\mathbb{Z}$.

(g) **True:** Take $G = \mathbb{Z}$ and $H = 2\mathbb{Z}$. We have that H is a cyclic group itself because every subgroup of a cyclic group is cyclic. Now since H is an infinite cyclic group, it must be isomorphic to $\mathbb{Z}$, by the Isomorphism Theorem of Cyclic Groups.

2. Consider an arbitrary group G and let $a, b \in G$.
- (a) Use induction to prove that $b(ab)^{n-1}a = (ba)^n$ for all $n \geq 1$.
 - (b) Suppose now that $(ab)^m = 1$ for some $m \geq 1$. Use (a) to show that $(ba)^m = 1$.

Solution: (a) For $n \geq 1$, let p_n be the statement $b(ab)^{n-1}a = (ba)^n$. It is clear that p_1 is true, since $b(ab)^{1-1}a = ba = (ba)^1$. Suppose now that p_k is true, that is, $b(ab)^{k-1}a = (ba)^k$ and let us show that p_{k+1} is also true. In fact:

$$b(ab)^k a = b(ab)(ab)^{k-1}a = (ba) (b(ab)^{k-1}a) \stackrel{\text{(I.H)}}{=} (ba)(ba)^k = (ba)^{k+1}.$$

By the Principle of Mathematical Induction, we obtain that p_n is true for all $n \geq 1$.

(b) Using that $(ab)^m = 1$ together with (a) we have that

$$(ba)^{m+1} = b(ab)^m a = ba.$$

Applying the Cancellation law, we obtain that $(ba)^m = 1$, as desired.

3. Let a and b elements in a group G such that they commute, i.e., $ab = ba$. Prove that a^{-1} and b^{-1} commute.

Solution: Multiplying $ab = ba$ on the right by a^{-1} gives $(ab)a^{-1} = (ba)a^{-1}$, that is, $aba^{-1} = b$. Multiplying again by a^{-1} but this time on the left produces $ba^{-1} = a^{-1}b$. Repeating the same argument with b gives $ba^{-1}b^{-1} = a^{-1}$ and $a^{-1}b^{-1} = b^{-1}a^{-1}$, which shows that a^{-1} and b^{-1} commute.

4. Let G be a group and $a, b \in G$ arbitrary elements.

(a) (10 marks) Prove (by induction on n) that

$$(aba^{-1})^n = ab^n a^{-1}, \text{ for every integer } n \geq 1.$$

(b) (5 marks) Prove that $(aba^{-1})^n = ab^n a^{-1}$ for all integers n .

Solution:

(a) Let p_n be the statement $(aba^{-1})^n = ab^n a^{-1}$. It is clear that p_1 holds since by definition $(aba^{-1})^1 = aba^{-1}$ and $ab^1 a^{-1} = aba^{-1}$. Next, assume that p_k is true, namely $(aba^{-1})^k = ab^k a^{-1}$, for some $k \geq 1$ and let us show the validity of p_{k+1} . By the definition of the powers of elements in a group we have that $(aba^{-1})^{k+1} = (aba^{-1})^k(aba^{-1})$. Applying p_k we obtain that

$$(aba^{-1})^{k+1} = (aba^{-1})^k(aba^{-1}) = (ab^k a^{-1})(aba^{-1}) = ab^k a^{-1} aba^{-1} = ab^{k+1} a^{-1},$$

showing p_{k+1} . Therefore, by the General Principle of Mathematical Induction, we can conclude that p_n is true for every integer $n \geq 0$.

(b) By definition of powers we have that $(aba^{-1})^0 = 1$ and $b^0 = 1$. So

$$(aba^{-1})^0 = 1 = aa^{-1} = ab^0 a^{-1}.$$

Let n be a negative integer. Then $-n$ is a positive integer and by (a) we have that $(aba^{-1})^{-n} = ab^{-n} a^{-1}$. On the other hand we have that

$$(aba^{-1})^n = ((aba^{-1})^{-n})^{-1} = (ab^{-n} a^{-1})^{-1} = (a^{-1})^{-1} (b^{-n})^{-1} a^{-1} = ab^n a^{-1},$$

finishing the proof.

5. Let H and K be subgroups of a group G . Consider the subset HK of G defined as follows:

$$HK = \{hk \mid h \in H, k \in K\}$$

Prove that HK is a subgroup of G if and only if $KH \subseteq HK$; notice that KH is defined exactly in the same way: $KH = \{kh \mid h \in H, k \in K\}$.

Solution: Assume first that HK is a subgroup of G and take $kh \in KH$, where $k \in K$ and $h \in H$. Our goal is to show that $kh \in HK$. Since both K and H are subgroups, we have that $k^{-1} \in K$ and $h^{-1} \in H$. By the definition of HK we have that $h^{-1}k^{-1} \in HK$, and since we are assuming that HK is a subgroup, we have that $(h^{-1}k^{-1})^{-1} \in HK$. Thus:

$$(h^{-1}k^{-1})^{-1} = (k^{-1})^{-1}(h^{-1})^{-1} = kh \in HK,$$

as desired.

Suppose now that $KH \subseteq HK$ and let us prove that HK is a subgroup of G . We will use the Subgroup Test. First notice that $1 = 1 \cdot 1 \in HK$. Next given $h_1k_1, h_2k_2 \in HK$ with $k_1, k_2 \in K$ and $h_1, h_2 \in H$ we have that

$$(h_1k_1)(h_2k_2) = h_1(k_1h_2)k_2, \tag{1}$$

but $k_1h_2 \in KH \subseteq HK$, and so $k_1h_2 = h'_2k'_1$ for some $k'_1 \in K$ and $h'_2 \in H$. Thus from (1) we get that

$$(h_1k_1)(h_2k_2) = h_1(k_1h_2)k_2 = h_1(h'_2k'_1)k_2 = (h_1h'_2)(k'_1k_2) \in HK.$$

To finish notice that

$$(h_1k_1)^{-1} = k_1^{-1}h_1^{-1} \in KH \subseteq HK.$$

The Subgroup Test allows us to conclude that HK is a subgroup of G .

6. Let G be a group and $g \in G$. Prove that $\langle g \rangle = \langle g^{-1} \rangle$.

Hint: Show that $\text{o}(g) = \text{o}(g^{-1})$.

Solution: It is clear that $(g^{-1})^k = (g^k)^{-1}$ for all $k \in \mathbb{Z}$. From here we obtain that $(g^{-1})^k = 1$ if and only if $g^k = 1$, proving that $\text{o}(g) = \text{o}(g^{-1})$.

The considerations above imply that $\langle g \rangle = \langle g^{-1} \rangle$.

7. Find all generators of

(a) $G = \langle g \rangle$, where $\text{o}(g) = \infty$.

(b) $\mathbb{Z}_{16}$.

(c) $G = \langle g \rangle$, if $\text{o}(g) = 5$.

Solution: (a) From the previous problem we know that g^{-1} is also a generator of G . We claim that the only generators of G are indeed g and g^{-1} . Suppose that g^m is another generator of G , that is, $G = \langle g^m \rangle$. Then $g = (g^m)^k = g^{mk}$ for some integer k . Applying that $\text{o}(g) = \infty$ we get that $mk = 1$. But m and k are integers so necessarily $m = \pm 1$.

(b) The generators of the additive cyclic group $\mathbb{Z}_{16} = \langle \bar{1} \rangle$ are the elements $k\bar{1} = \bar{k} \in \mathbb{Z}_{16}$, where $\text{gcd}(k, 16) = 1$. Hence, $\mathbb{Z}_{16}$ has generators $\bar{1}, \bar{3}, \bar{5}, \bar{7}, \bar{9}, \bar{11}, \bar{13}, \bar{15}$.

(c) Recall that if $G = \langle g \rangle$ with $\text{o}(g) = n$, then g^k generates G if and only if $\text{gcd}(k, n) = 1$. In our case, $n = 5$ and the positive integers k such that $\text{gcd}(k, 5) = 1$ are 1, 2, 3, 4. Thus, the generators of G are g, g^2, g^3, g^4 .

8. (a) Determine all the elements of the multiplicative groups $\mathbb{Z}_{14}^*$ and $\mathbb{Z}_{18}^*$.
 (b) Prove that $\mathbb{Z}_{14}^*$ and $\mathbb{Z}_{18}^*$ are both cyclic groups of order 6. Conclude that there are isomorphic.

Recall that $\mathbb{Z}_n^*$ denotes the subset of $\mathbb{Z}_n$ consisting on the elements that have a multiplicative inverse.

Solution: (a) By definition $\mathbb{Z}_{14}^*$ (respectively, $\mathbb{Z}_{18}^*$) consists on the elements of the form $\bar{a}$, such that $\gcd(a, 14) = 1$ (respectively, $\gcd(a, 18) = 1$). From here we obtain that

$$\mathbb{Z}_{14}^* = \{\bar{1}, \bar{3}, \bar{5}, \bar{9}, \bar{11}, \bar{13}\}, \quad \mathbb{Z}_{18}^* = \{\bar{1}, \bar{5}, \bar{7}, \bar{11}, \bar{13}, \bar{17}\}.$$

(b) To prove that $\mathbb{Z}_{14}^*$ is a cyclic group of order 6, we will show that $\bar{3} \in \mathbb{Z}_{14}^*$ is a generator. We compute the powers of $\bar{3}$:

$$\begin{aligned} \bar{3}^0 &= \bar{1}, \\ \bar{3}^1 &= \bar{3}, \\ \bar{3}^2 &= \bar{9}, \\ \bar{3}^3 &= \overline{27} = \bar{13}, \\ \bar{3}^4 &= \overline{39} = \bar{11}, \\ \bar{3}^5 &= \bar{5}, \\ \bar{3}^6 &= \overline{15} = \bar{1} \end{aligned}$$

We can see that every element of $\mathbb{Z}_{14}^*$ is a power of $\bar{3}$. Thus $\mathbb{Z}_{14}^*$ is a cyclic group of order 6, and so $\mathbb{Z}_{14}^* \cong \mathbb{Z}_6$. In a similar way, you can show that $\mathbb{Z}_{18}^*$ is a cyclic group of order 6 generated by $\bar{5}$ (complete the details), and so $\mathbb{Z}_{18}^* \cong \mathbb{Z}_6$. Therefore $\mathbb{Z}_{14}^* \cong \mathbb{Z}_{18}^* \cong \mathbb{Z}_6$.

9. Consider the additive cyclic group $\mathbb{Z}_{48}$.

- (a) Determine all the subgroups of $\mathbb{Z}_{48}$; list all the elements of each subgroup.
- (b) Draw the lattice diagram of all the subgroups $\mathbb{Z}_{48}$.

Solution: (a). The positive divisors of 48 are 1, 2, 3, 4, 6, 8, 12, 16, 24, 48. Thus, the Fundamental Theorem of Finite Cyclic Groups tells us that $\mathbb{Z}_{48}$ has exactly 10 subgroups, exactly one per positive divisor of 48. It is clear that the subgroups of orders 1 and 48 are $\{0\}$ and $\mathbb{Z}_{48}$, respectively. Moreover, for $k \in \{2, 3, 4, 6, 8, 12, 16, 24\}$, applying the Fundamental Theorem of Finite Cyclic Groups we obtain that $\langle \frac{48}{k} \bar{1} \rangle$ is the unique subgroup of $\mathbb{Z}_{48}$ of order k .

Using this, we obtain that

$$\text{order 2: } \left\langle \frac{48}{2} \bar{1} \right\rangle = \langle \bar{24} \rangle = \{0, \bar{24}\},$$

$$\text{order 3: } \left\langle \frac{48}{3} \bar{1} \right\rangle = \langle \bar{16} \rangle = \{0, \bar{16}, \bar{32}\},$$

$$\text{order 4: } \left\langle \frac{48}{4} \bar{1} \right\rangle = \langle \bar{12} \rangle = \{0, \bar{12}, \bar{24}, \bar{36}\},$$

$$\text{order 6: } \left\langle \frac{48}{6} \bar{1} \right\rangle = \langle \bar{8} \rangle = \{0, \bar{8}, \bar{16}, \bar{24}, \bar{32}, \bar{40}\},$$

$$\text{order 8: } \left\langle \frac{48}{8} \bar{1} \right\rangle = \langle \bar{6} \rangle = \{0, \bar{6}, \bar{12}, \bar{18}, \bar{24}, \bar{30}, \bar{36}, \bar{42}\},$$

$$\text{order 12: } \left\langle \frac{48}{12} \bar{1} \right\rangle = \langle \bar{4} \rangle = \{0, \bar{4}, \bar{8}, \bar{12}, \bar{16}, \bar{20}, \bar{24}, \bar{28}, \bar{32}, \bar{36}, \bar{40}, \bar{44}\},$$

$$\text{order 16: } \left\langle \frac{48}{16} \bar{1} \right\rangle = \langle \bar{3} \rangle = \{0, \bar{3}, \bar{6}, \bar{9}, \bar{12}, \bar{15}, \bar{18}, \bar{21}, \bar{24}, \bar{27}, \bar{30}, \bar{33}, \bar{36}, \bar{39}, \bar{42}, \bar{45}\},$$

$$\begin{aligned} \text{order 24: } \left\langle \frac{48}{24} \bar{1} \right\rangle &= \langle \bar{2} \rangle \\ &= \{0, \bar{2}, \bar{4}, \bar{6}, \bar{8}, \bar{10}, \bar{12}, \bar{14}, \bar{16}, \bar{18}, \bar{20}, \bar{22}, \bar{24}, \bar{26}, \bar{28}, \bar{30}, \bar{32}, \bar{34}, \bar{36}, \bar{38}, \bar{40}, \bar{42}, \bar{44}, \bar{46}\}. \end{aligned}$$

(b) Notice that

$$\begin{aligned} \langle \bar{24} \rangle &\subseteq \langle \bar{2} \rangle, & \langle \bar{24} \rangle &\subseteq \langle \bar{3} \rangle, & \langle \bar{24} \rangle &\subseteq \langle \bar{4} \rangle, & \langle \bar{24} \rangle &\subseteq \langle \bar{6} \rangle & \text{ and } & \langle \bar{24} \rangle &\subseteq \langle \bar{8} \rangle, \\ \langle \bar{16} \rangle &\subseteq \langle \bar{8} \rangle, & \langle \bar{16} \rangle &\subseteq \langle \bar{4} \rangle & \text{ and } & \langle \bar{16} \rangle &\subseteq \langle \bar{2} \rangle, \\ \langle \bar{12} \rangle &\subseteq \langle \bar{6} \rangle, & \langle \bar{12} \rangle &\subseteq \langle \bar{4} \rangle & \text{ and } & \langle \bar{12} \rangle &\subseteq \langle \bar{2} \rangle, \\ \langle \bar{8} \rangle &\subseteq \langle \bar{4} \rangle & \text{ and } & \langle \bar{8} \rangle &\subseteq \langle \bar{2} \rangle, \\ \langle \bar{6} \rangle &\subseteq \langle \bar{3} \rangle & \text{ and } & \langle \bar{6} \rangle &\subseteq \langle \bar{2} \rangle, \\ \langle \bar{4} \rangle &\subseteq \langle \bar{2} \rangle. \end{aligned}$$

Using these we get the following lattice diagram:

